

Physical AI in the Operating Room: A Genesis Mission Robotics Testbed

A Phase 1, first-in-human, LLM-advised eight-arm robotic Whipple with perioperative daraxonrasib (RMC-6236) in KRAS-mutated PDAC, instrumented as a closed-loop, openly formatted, verifiable experiment.

EO 14363

Gold Standard Science

Open Data Formats

5 Years

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Submitted as an independent scientist under the funding approach set out in Science: A New Golden Age (July 2026). Not medical or regulatory advice, and not endorsed by the FDA, NIH, HHS, DOE, an IRB, ICH, or any sponsor. UC San Diego is named as the intended partner of choice and is not a sponsor, site, or endorser.

1 The Robotics Mission Names This Work

The FY 2028 budget priorities memorandum annexed to *Science: A New Golden Age* names six national missions, as depicted in Table 1 [1]. The Robotics mission is written as “general-purpose autonomous systems capable of dexterous manipulation, mobility, and reliable operation in real-world environments, to initiate the era of physical AI-driven scientific discovery and American reindustrialization.” An eight-arm robotic pancreaticoduodenectomy under a governed advisory model is that sentence applied to a human operating room, which is the least forgiving real-world environment available.

The same memorandum asks agencies to realign a roughly **\$200 billion** annual portfolio toward the individual scientist, to fund AI as a new instrument of discovery rather than for incremental gains, and to use federal purchasing power to drive instrument redesign toward open interfaces, standardized data formats, and cross-vendor interoperability [2]. This proposal is to be the small, hard case that tests all three at once.

Annex directive	What this testbed contributes
Invest in robotics, automated laboratories, and closed-loop AI workflows	A closed loop whose actuator is a surgical robot and whose gate is a licensed human
Drive instrument redesign toward open interfaces and standardized formats	An open advisory logging schema and an open stop-latency measurement method
Gold Standard Science: reproducibility, negative results, error and uncertainty	Escalation data released with negative results and an uncertainty statement

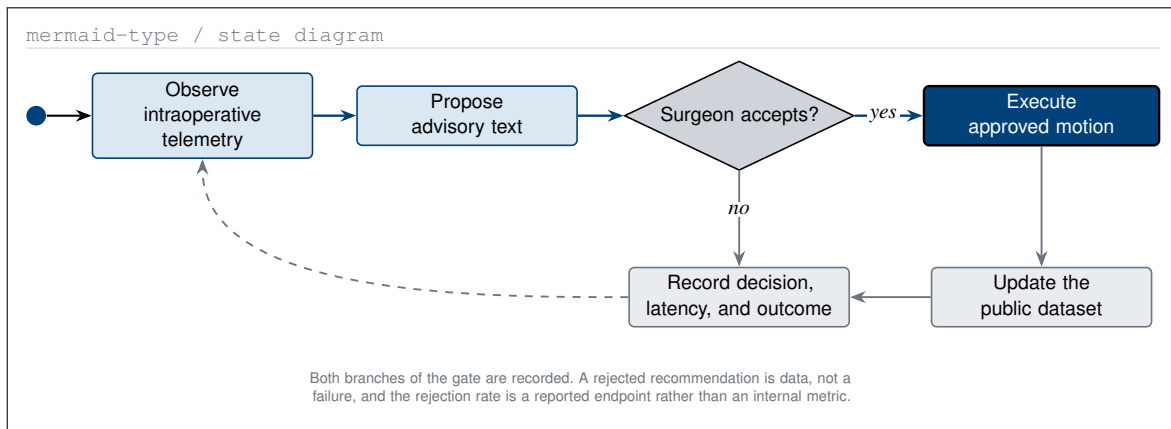
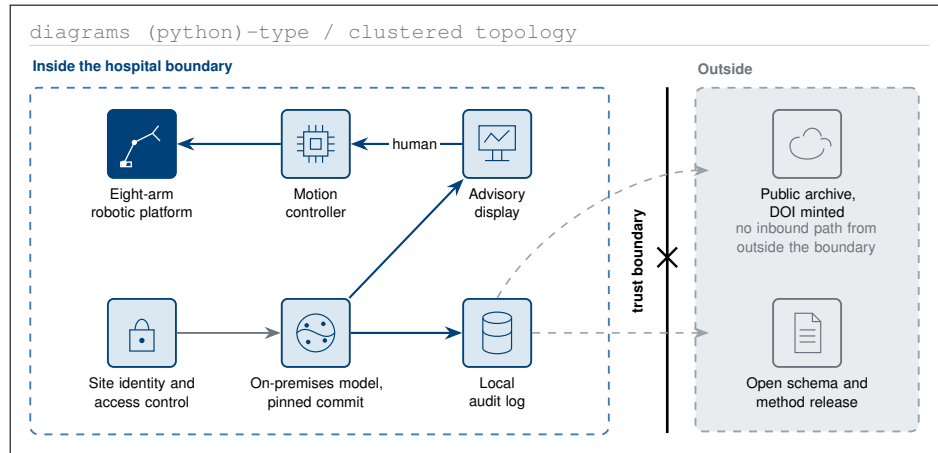
Table 1. Three directives from the annexed FY 2028 memorandum, and the specific artifact this testbed contributes to each. Every artifact is released, not merely reported.

2 What Runs Where

The Genesis Mission’s data infrastructure work assumes instruments that emit open, curated records. This program’s contribution is a topology in which the model is useful and still cannot act: everything that could influence a motion sits inside the hospital boundary, and nothing that leaves it can return, seen in Figure 1.

3 A Closed Loop with a Human Gate

The report calls closed-loop autonomous experimentation a strategic imperative and notes that materials work taking around twenty years from laboratory to deployment could be compressed by an order of magnitude [2]. A clinical loop cannot be autonomous stated in Figure 2, and this proposal doesn’t ask for that. It asks for the loop to be closed everywhere it can be, with one deliberate human gate that is measured.



4 Gold Standard Science, Artifact by Artifact

5 Mission Alignment, Budget, and the Partner Site

The request is **\$700,000 per year for five years**, \$3,500,000 total, with no cost share (Figure 4) [9]. The operation is a staged eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib in resectable or borderline-resectable KRAS G12-mutated PDAC, open-label, single-arm, 3+3 escalation, up to 18 treated participants [10, 11]. Arm-level stop is specified at 3 milliseconds and system-wide stop at 500 milliseconds, measured on the bench and re-measured at each cohort boundary.

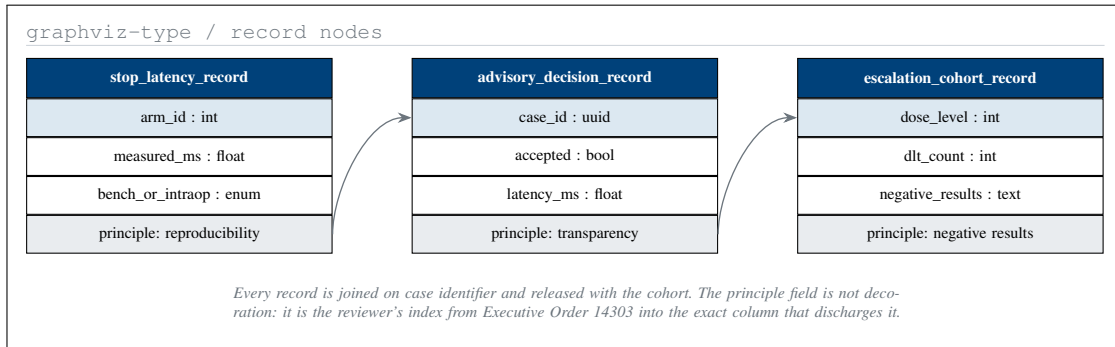


Figure 3. Three verification records, field by field, each carrying the Gold Standard Science principle it discharges. The join is on case identifier, and all three are released with the cohort.

d2-type / mission alignment grid

Genesis Mission challenge	What this testbed supplies	Released in
Problem selection	A named disease with a published protocol already in hand	Year 1
Institutional capacity	One independent performer contracting one qualified site	Year 1
Data infrastructure	Open advisory logging schema and stop-latency method	Years 2 and 3
AI with experimental infrastructure	A closed loop whose actuator is a surgical robot	Years 3 to 5

Figure 4. The Genesis Mission's four named challenges, what this testbed supplies against each, and the year the corresponding artifact is released into the public record.

Moore's Cancer Center is the intended partner of choice, approached at the feasibility stage only. UC San Diego is not a partner, sponsor, site, or endorser, because no written authorization exists.

Positioning note. First in human refers to the integrated surgical and advisory workflow, not to daraxonrasib, which is investigational and already in Phase 3 evaluation. Robotic configuration is specified by manufacturer, model, cleared configuration, arm count, and software version at the site agreement, and no drug supply or letter of authorization is in place with agent's developer.

Scope of Claims

Claimed: a published protocol and IND, an audited simulation package scoring 81.9 under ASME V&V 40 and ICH M15, and a testbed design that releases its schema, its method, and its negative results. Not claimed: that the simulations validate RASolute 302, that the advisory system holds any regulatory clearance, that anything here has been used in a human, or that UC San Diego has agreed to anything. The trial does not exist until a site, IRB, active IND, monitoring, insurance, and a sponsor of record are all in place.

Availability and Conflicts

Every work cited to a DOI is openly deposited; code, verification notebooks, and figure sources are in the repository [12]. The applicant holds no equity in Revolution Medicines or any robotic surgery vendor, and no sponsor, contract research organization, or medical society reviewed this document.

References

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